

P 3.6-11 Find i and $R_{eq\ a-b}$ if $v_{ab} = 40\text{ V}$ in the circuit of Figure P 3.6-11.

Answer: $R_{eq\ a-b} = 8\ \Omega$, $i = 5/6\text{ A}$

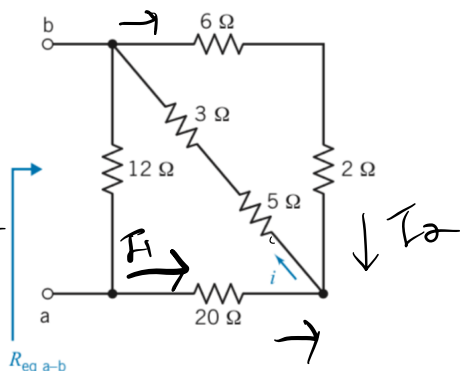
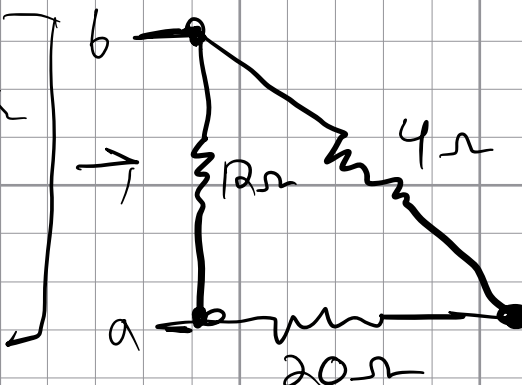


Figure P 3.6-11

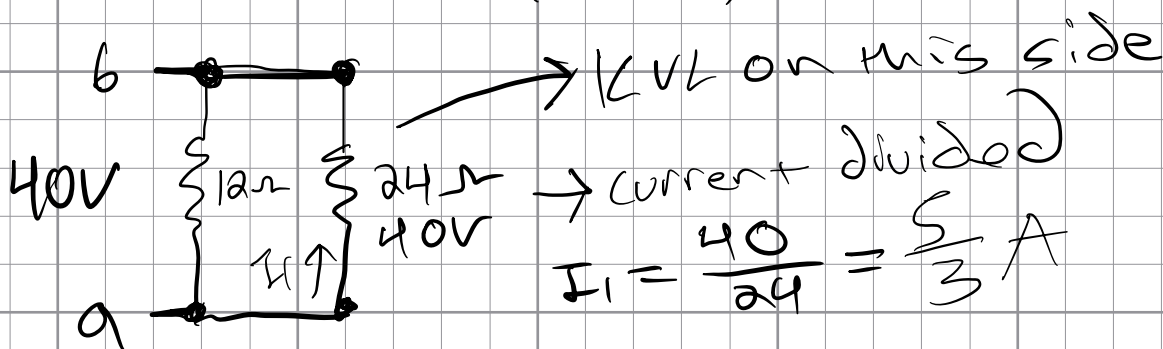
Ranvir Thind
05-09-26

1. $\frac{\text{Series}}{6+2=8\ \Omega}$ $\frac{\text{Series}}{3+5=8\ \Omega}$

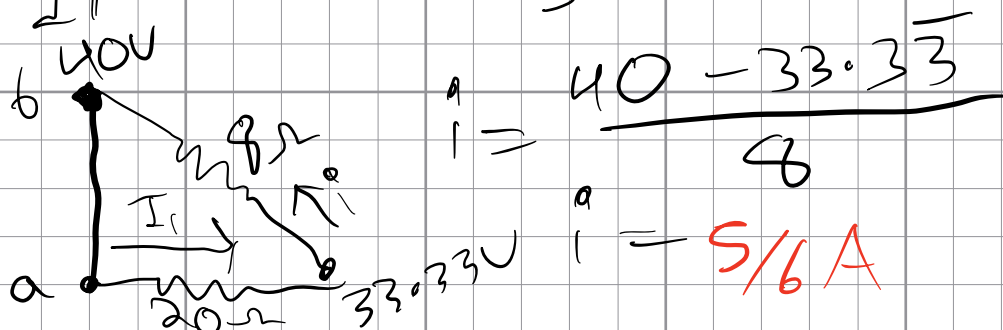
3. $\frac{\text{Parallel}}{\left(\frac{1}{8} + \frac{1}{8}\right)^{-1} = 4\ \Omega}$



4. $\frac{\text{Series}}{4+20=24\ \Omega}$ $\frac{\text{Parallel}}{R_{eq} = \left(\frac{1}{12} + \frac{1}{24}\right)^{-1} = 8\ \Omega}$



$I_1 = i - I_2 \Rightarrow \frac{5}{3} = i - I_2$



P 3.6-15 The circuit shown in Figure P 3.6-15 contains seven resistors, each having resistance R . The input to this circuit is the voltage source voltage v_s . The circuit has two outputs, v_a and v_b . Express each output as a function of the input.

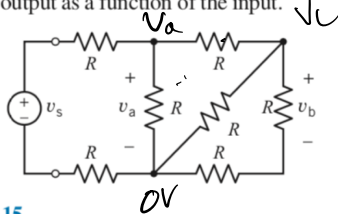
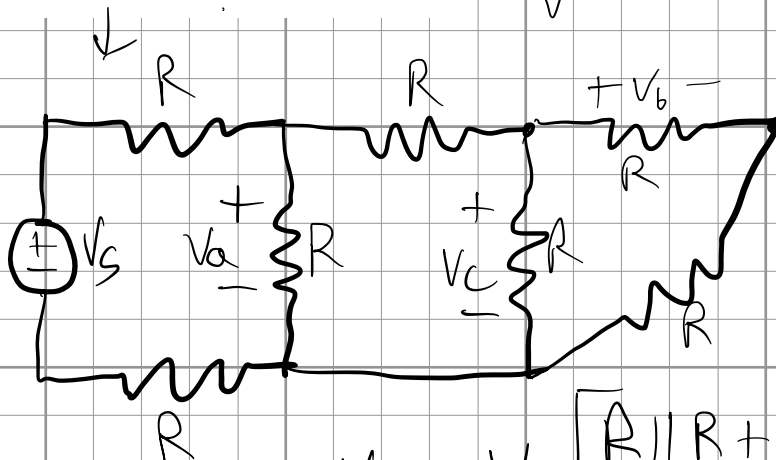


Figure P 3.6-15

$$\begin{matrix} v_a(V_s) \\ v_b(V_s) \end{matrix}$$

$$V_{out} = V_{in} \left(\frac{R_a}{R_a + R_1} \right)$$



$$V_a = V_s \frac{[R \parallel R + (R \parallel 2R)]}{2R + [R \parallel R + (R \parallel 2R)]}$$

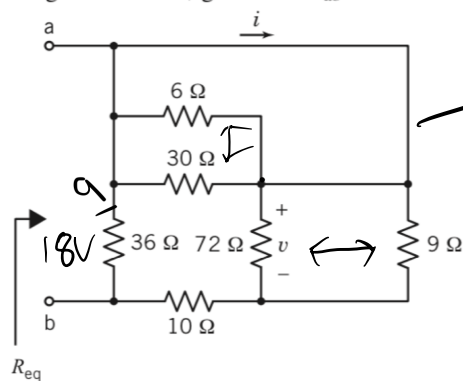
$$V_a = \frac{5}{21} V_s$$

$$V_c = V_a \frac{[R \parallel 2R]}{R + (R \parallel 2R)} = \frac{2}{5} V_a = \frac{2}{21} V_s$$

$$V_b = V_c \frac{R}{2R} = \frac{2}{21} V_s \left(\frac{1}{2} \right)$$

$$V_b = \frac{1}{21} V_s$$

P 3.6-20 Determine the values of i , v , and R_{eq} for the circuit shown in Figure P 3.6-20, given that $v_{ab} = 18$ V.



→ Parallel Short Circuit

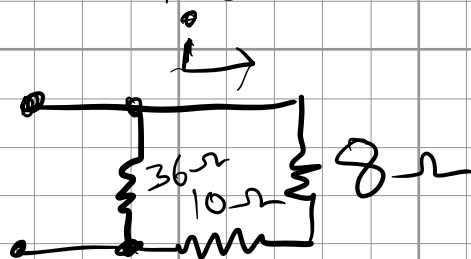


Figure P 3.6-20

$$V = 18 \frac{R_2}{R_1 + R_2} = \frac{\left(\frac{1}{72} + \frac{1}{9}\right)^{-1}}{10 + \left(\frac{1}{72} + \frac{1}{9}\right)^{-1}}$$

$$V = 18V \frac{8}{18} = 8V$$

$$R_{eq} = 10 + 8 \parallel 36 = 12\Omega$$

$$i = \frac{V}{R} = \frac{8}{8} = 1A$$

P 3.6-21 Determine the value of the resistance R in the circuit shown in Figure P 3.6-21, given that $R_{eq} = 9 \Omega$.

Answer: $R = 15 \Omega$

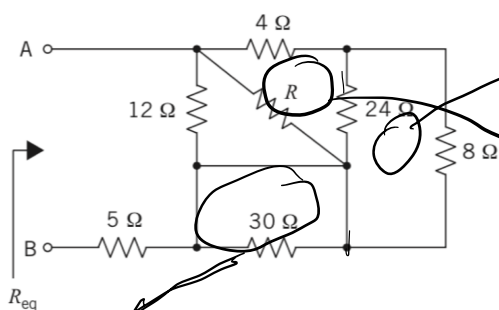


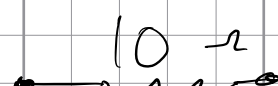
Figure P 3.6-21

Parallel

$$\left(\frac{1}{24} + \frac{1}{8} \right) = 6 \Omega$$

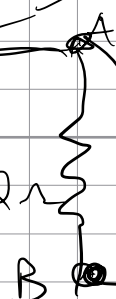
Series

$$6 + 4 = 10 \Omega$$



Parallel

$$\left(\frac{1}{10} + \frac{1}{R} \right)^{-1} = 12 \Omega$$



$$\frac{10R}{10+R}$$

Parallel

$$\left(\frac{1}{12} + \frac{10+R}{10R} \right)^{-1}$$

Short circuit || Resistor || Short circuit

Series

$$\frac{120R}{10R + 12R + 120} + 5 = R_{eq} = 9$$

$$120R = 88R + 480$$

$$R = 15 \Omega$$

P 3.6-21 Determine the value of the resistance R in the circuit shown in Figure P 3.6-21, given that $R_{eq} = 9\ \Omega$.

Answer: $R = 15\ \Omega$

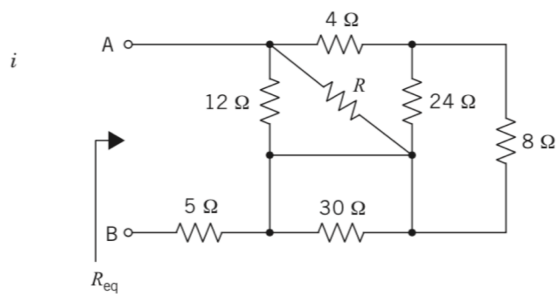


Figure P 3.6-21

Same as Last Problem

P 3.6-42 Determine the values of i_a , i_b , i_2 , and v_1 in the circuit shown in Figure P 3.6-42.

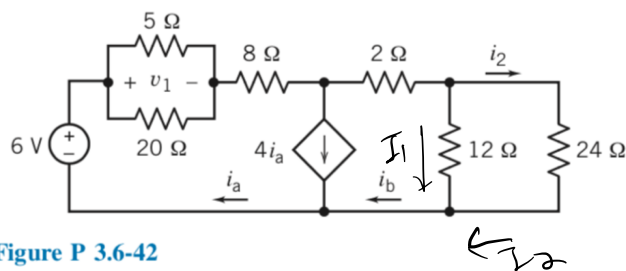
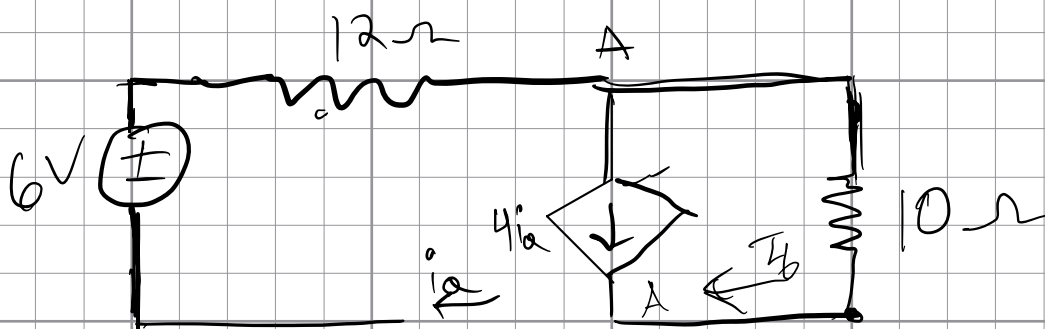


Figure P 3.6-42

1. Simplify/Rewrite circuit.

2. KVL + KCL



KCL @ A

$$I_b = -3i_a$$

KVL

$$-6 + 12i_a + 10i_b = 0$$

$$-18i_a = 6$$

$$-6 + 10 - \frac{8}{3} + v_1 = 0 \quad \begin{matrix} i_a = -\frac{1}{3} \text{ A} \\ i_b = 1 \text{ A} \end{matrix}$$

$$v_1 = -\frac{4}{3} \text{ V}$$

$$I_a = \frac{v}{R_{I_a}} = I_b \times \left(\frac{1}{12} + \frac{1}{24} \right)^{-1} / 24 \Omega$$

$$I_2 = \frac{1}{3} \text{ A}$$

P 3.6-47 Determine the values of the voltages v_1 and v_2 and of the current i_3 in the circuit shown in Figure P 3.6-47.

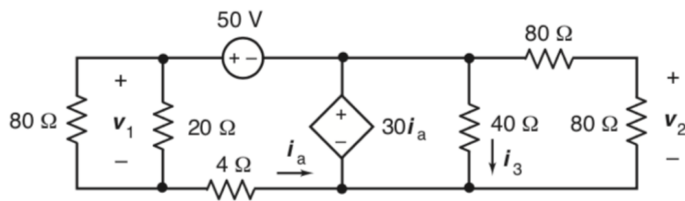
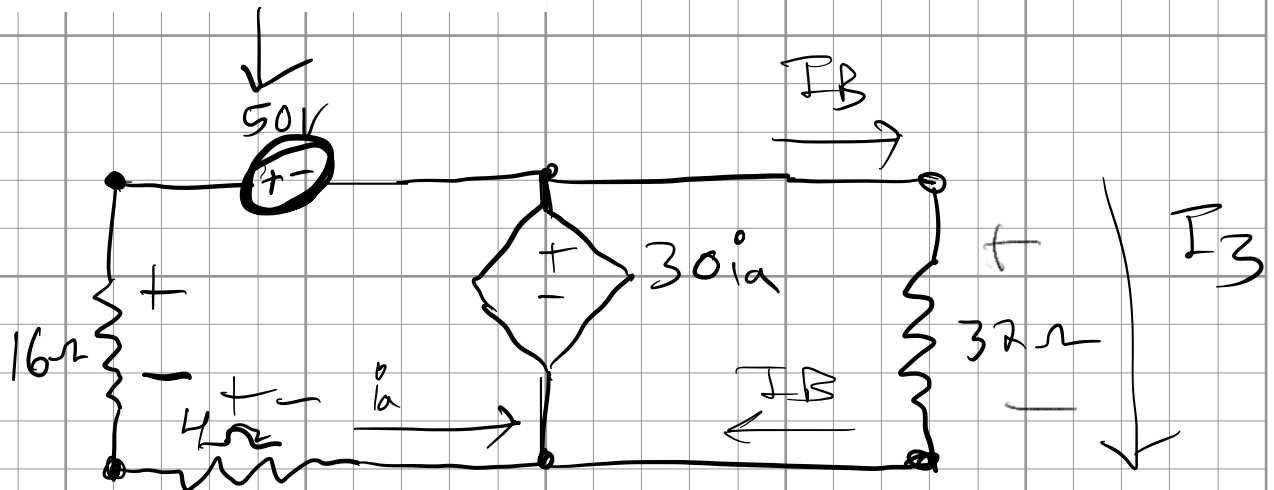


Figure P 3.6-47



Left Mesh KVL

$$-50 + 16i_a + 4i_a - 30i_a = 0$$

$$i_a = -5A \quad v_1 = 16 \times -5 = -80V$$

Right Mesh KVL

$$32I_B = 30i_a$$

$$I_B = -4.6875$$

$$v_2 = 30i_a \left(\frac{160}{40 + 160} \right) = -75V$$

$$I_3 = I_B \times \frac{160}{200} = -3.75V$$

P 3.6-23 Determine the values of r , the gain of the CCVS, and g , the gain of the VCCS, for the circuit shown in Figure P 3.6-23.

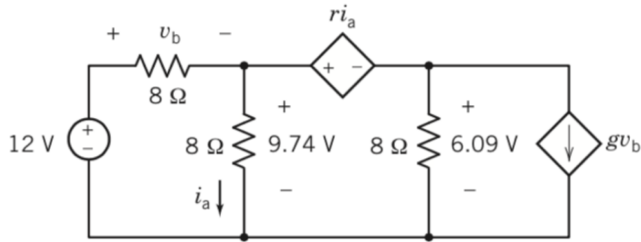


Figure P 3.6-23

$$i_a = 9.74 / 8 = 1.2175 \text{ A}$$

KVL $V_b = 12 - 9.74 \text{ V} = 2.26 \text{ V}$

$$r i_a = 9.74 - 6.09$$

$$r i_a = 3.65$$

$$r = 2.998 \text{ V/A}$$

KCL $g v_b + \frac{6.09}{8} - \frac{2.26}{8} + \frac{9.74}{8}$

$$g v_b = -1.696 \text{ A}$$

$$g = \frac{g v_b}{2.26 \text{ V}} = -0.75$$

DP 3-3 A phonograph pickup, stereo amplifier, and speaker are shown in Figure DP 3-3a and redrawn as a circuit model as shown in Figure DP 3-3b. Determine the resistance R so that the voltage v across the speaker is 16 V. Determine the power delivered to the speaker.

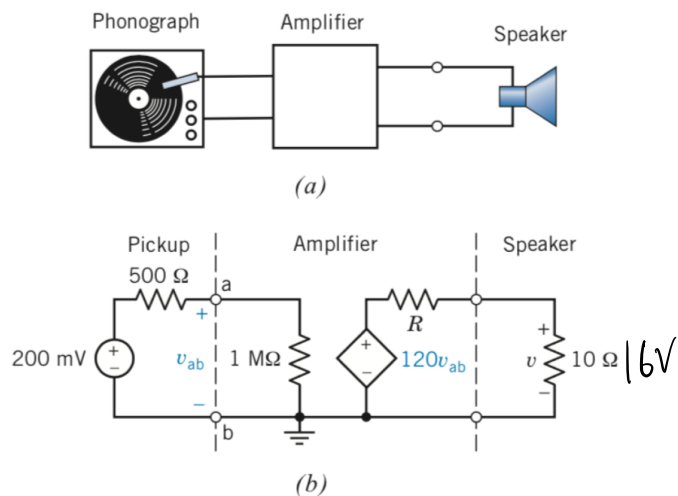


Figure DP 3-3 A phonograph stereo system.

$$V_{out} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$V_{ab} = 200 \text{ mV} \left(\frac{1 \text{ M}}{1 \text{ M} + 500} \right) = 199.9 \text{ mV}$$

$$V = 120 V_{ab} \left(\frac{10}{10 + R} \right)$$

$$16 = 120 \times 0.2 \left(\frac{10}{10 + R} \right)$$

$$R = 5 \Omega \quad I = \frac{16}{10} = 1.6$$

$$P_{5\Omega} = 1.6^2 \times 10 = 25.6 \text{ W}$$